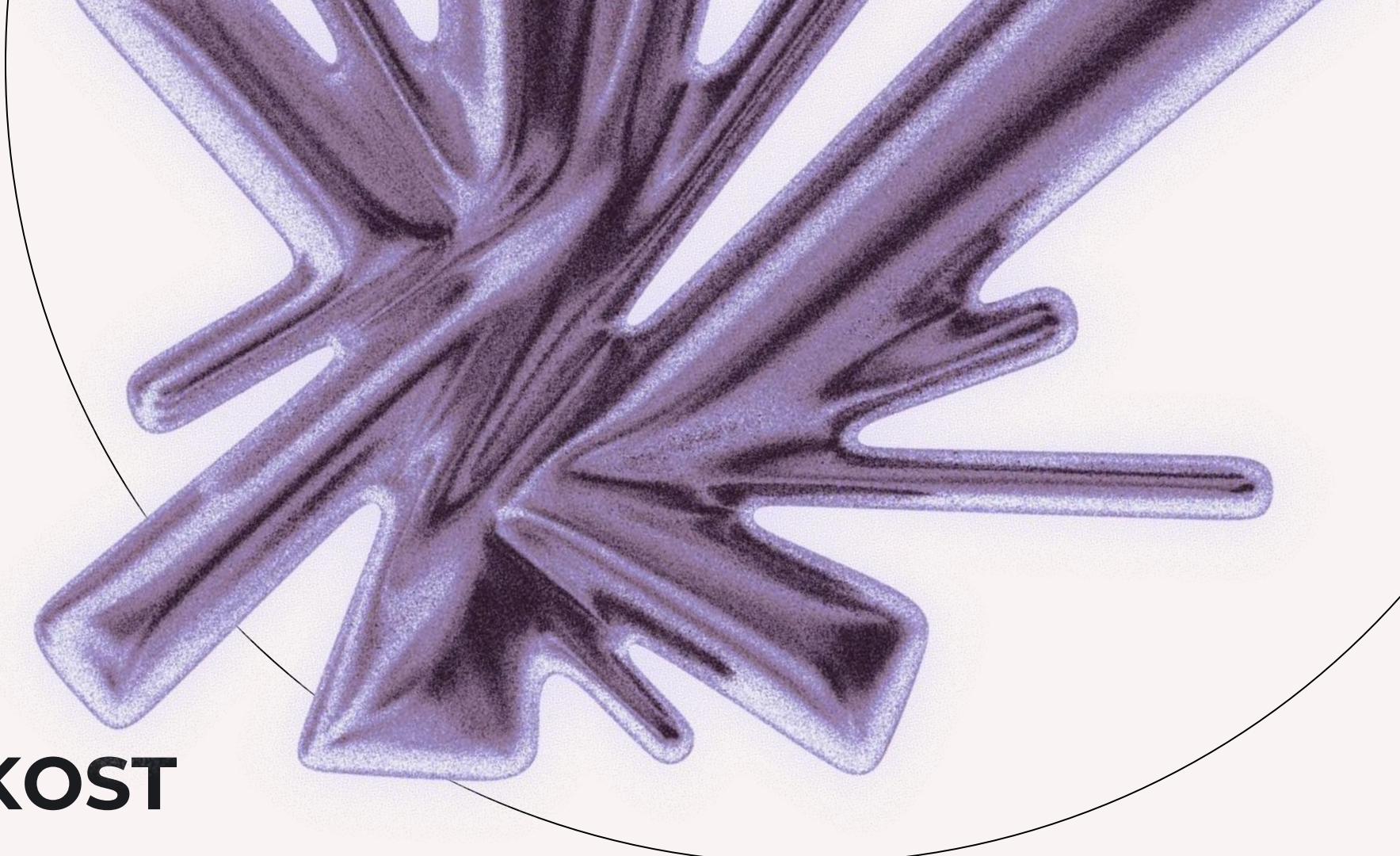
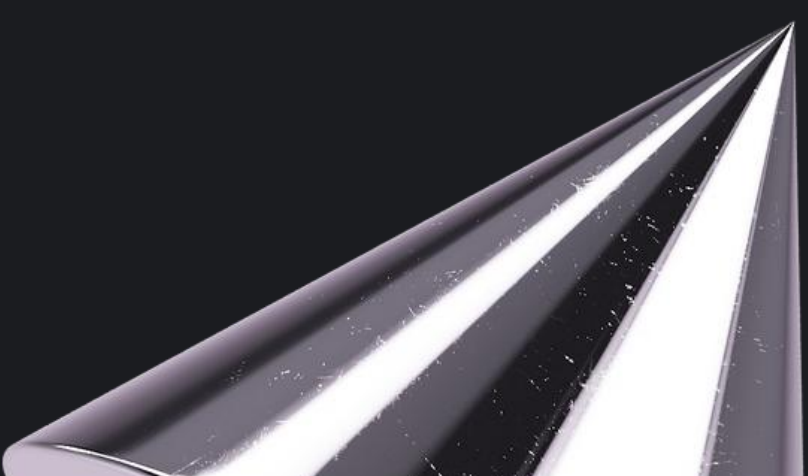




PAPER PRESENTATION BY YAROSLAVA KOST

# Final Project

## *Terrain Generation*

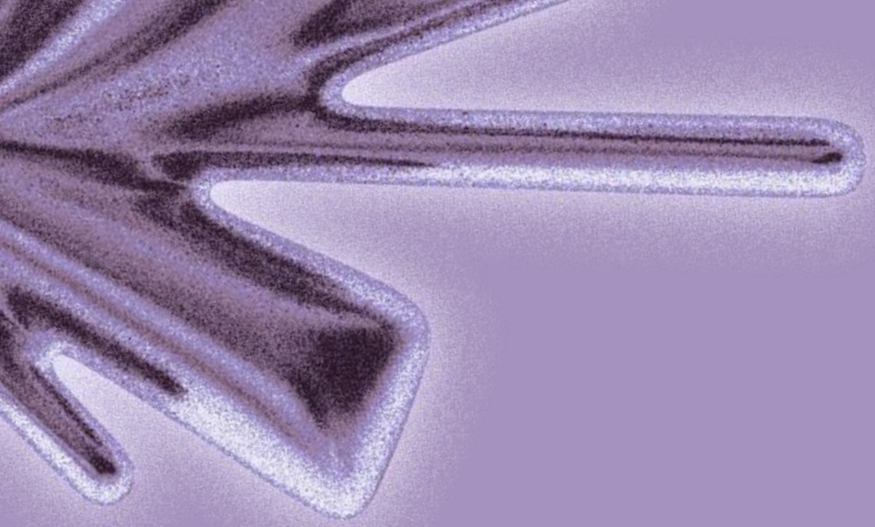




# Terrains







# My Tool

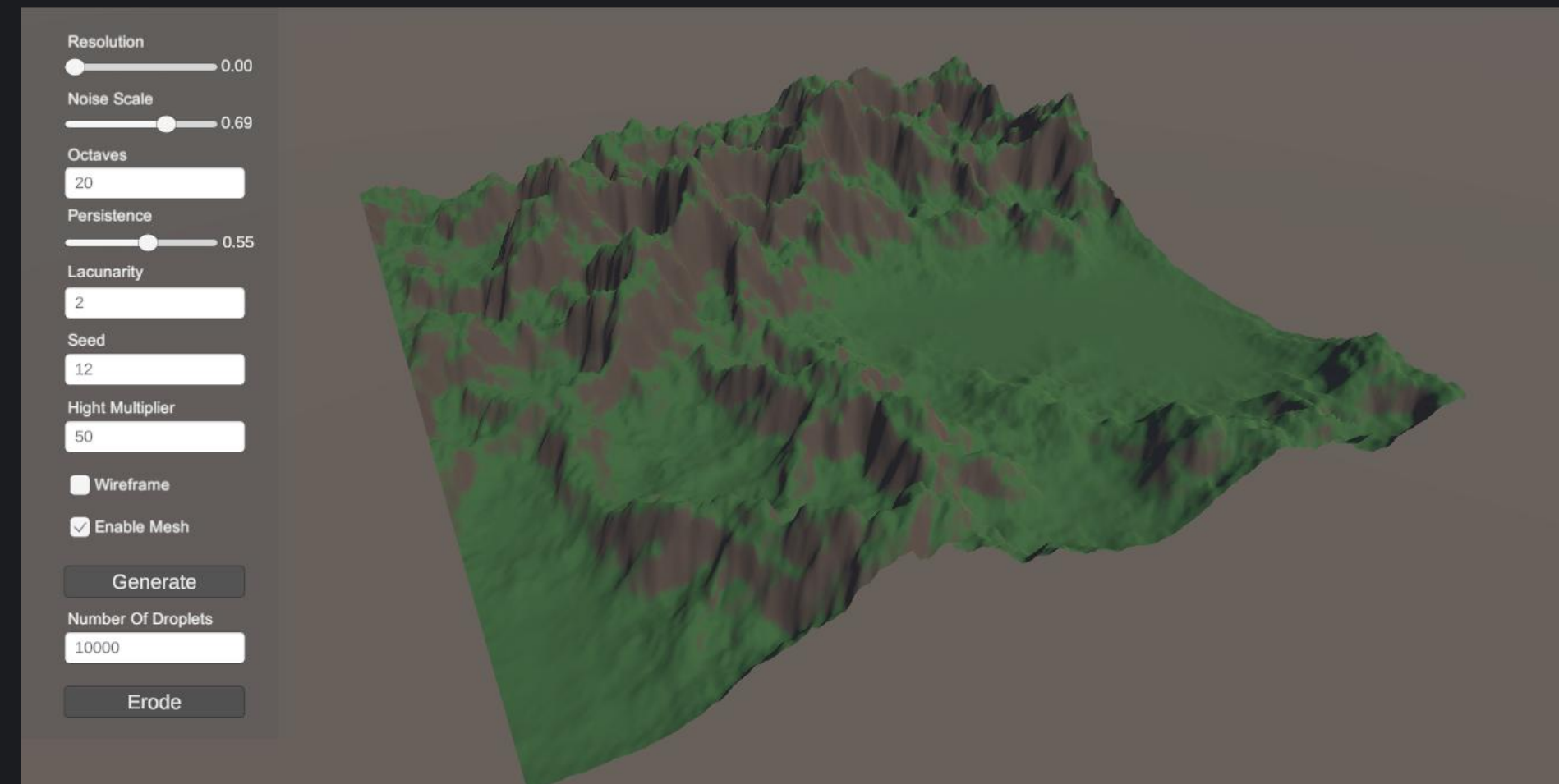
- Perlin Noise + fractal Brownian motion
- Hydraulic erosion simulation
- Adjustable parameters with intuitive UI

Result: Realistic Terrain!

# My Tool

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# Erosion

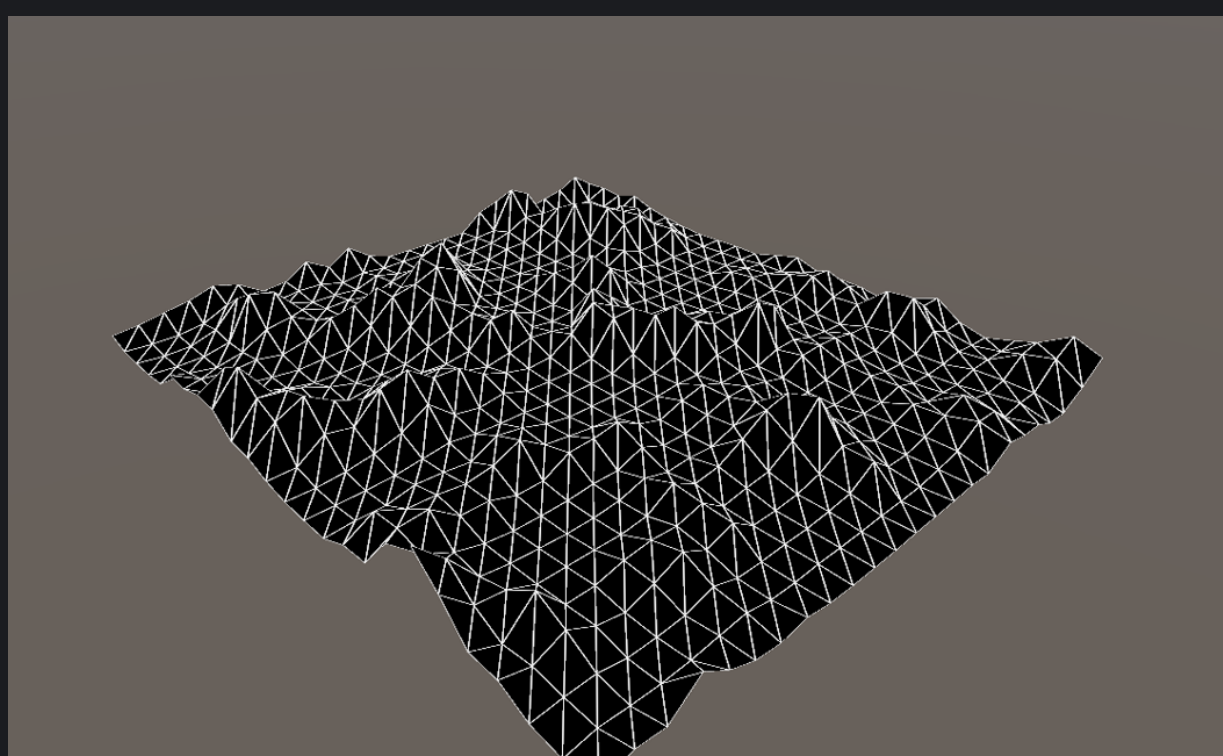
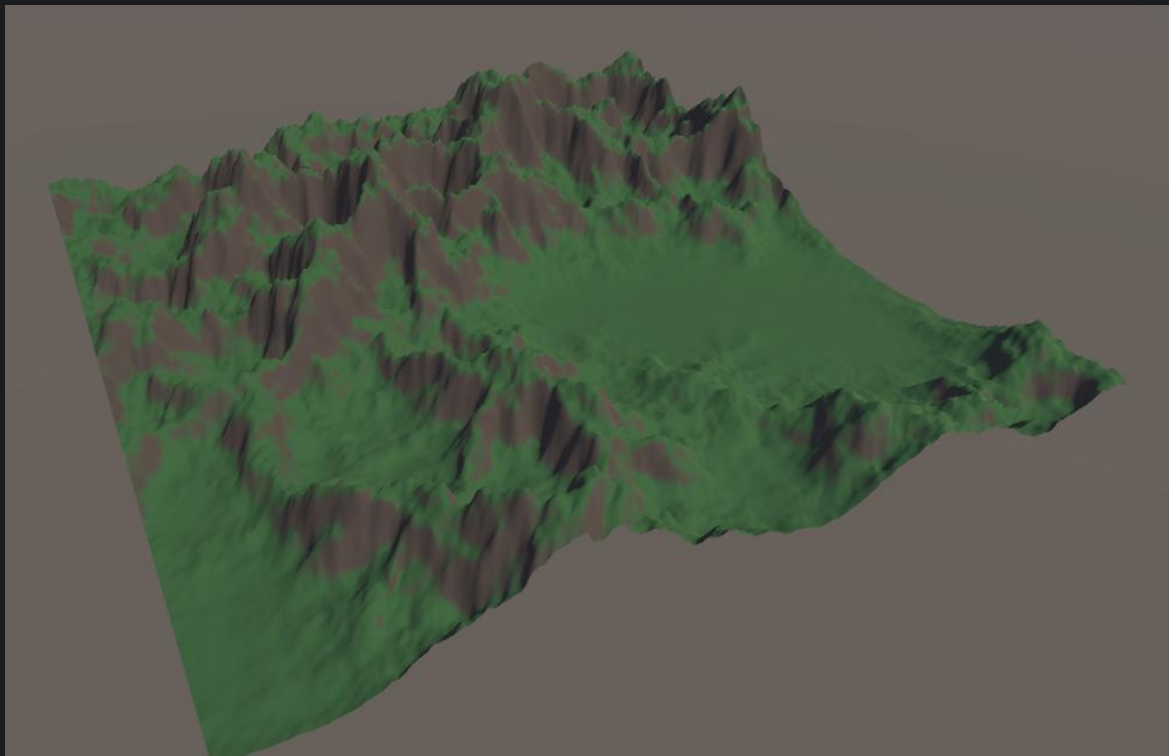
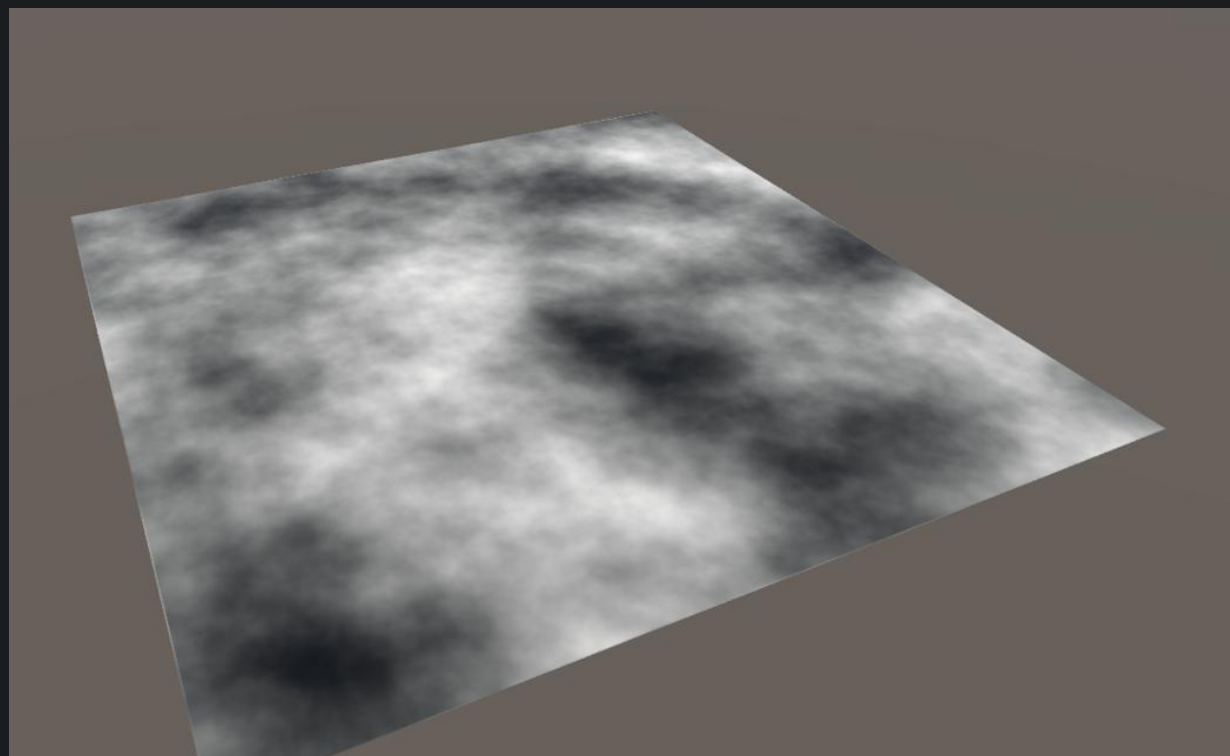
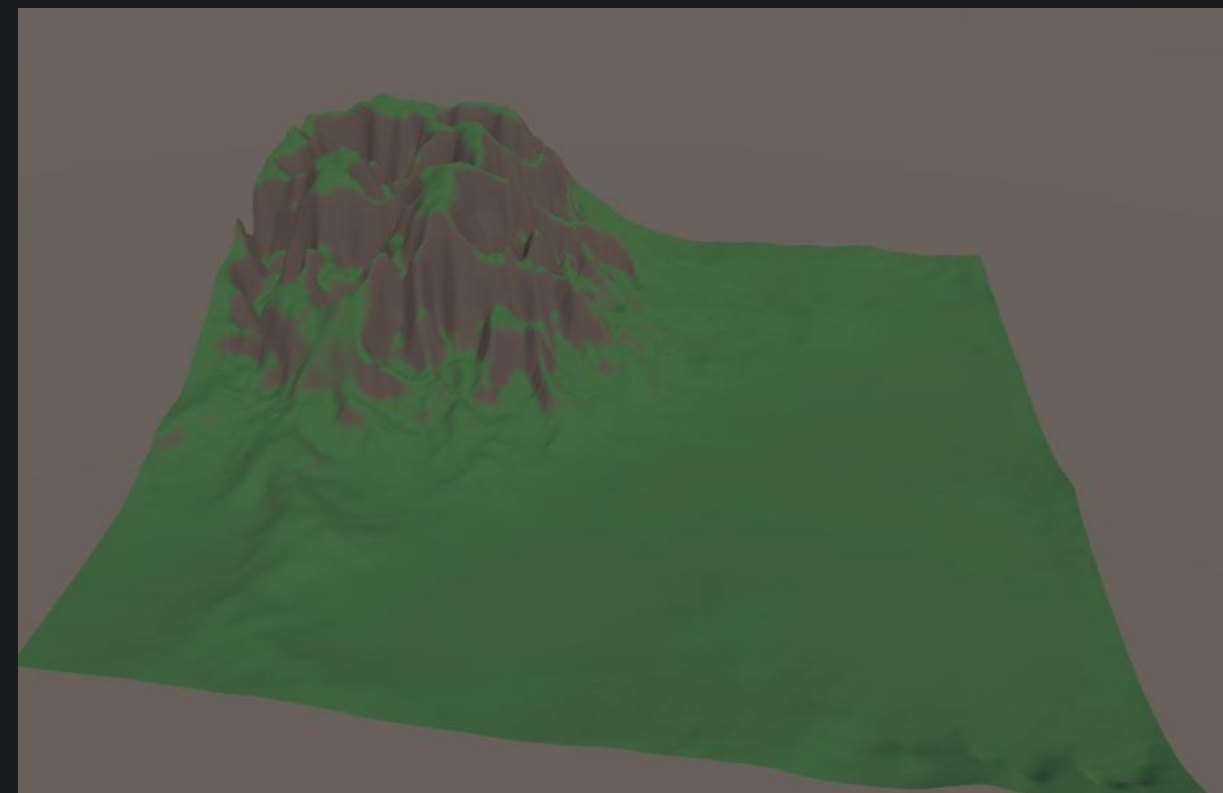
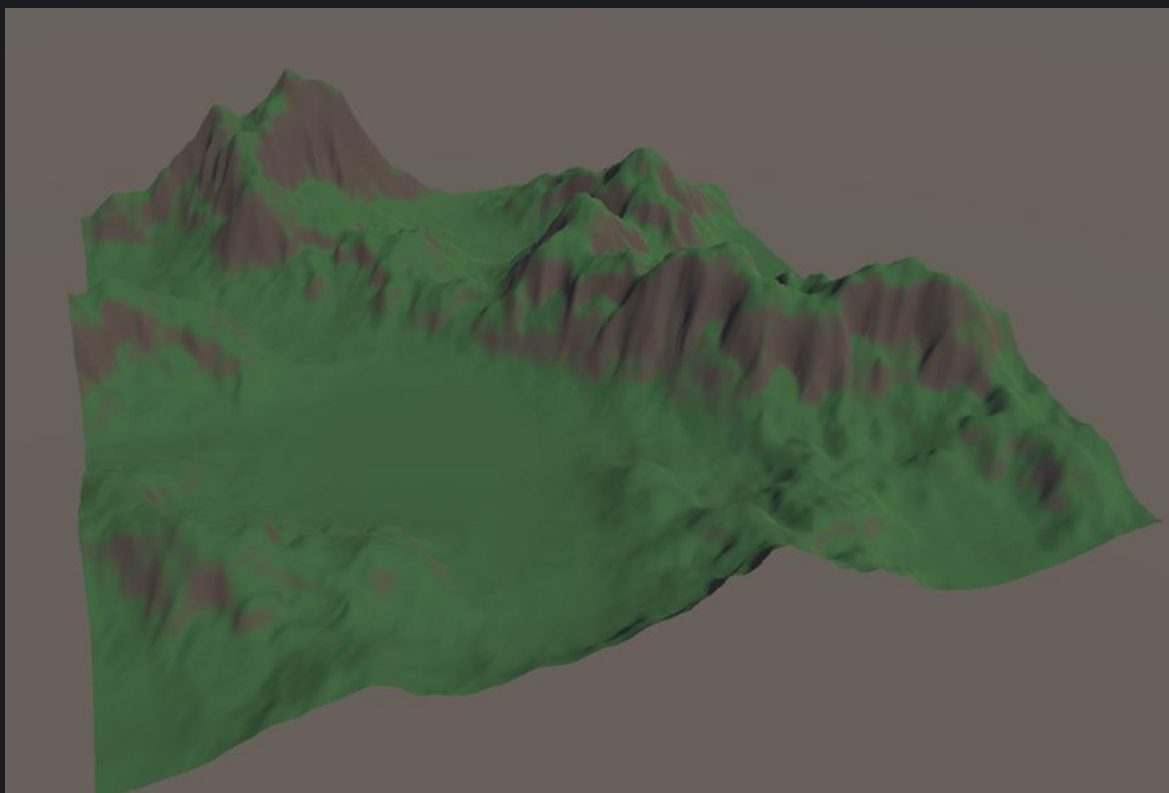
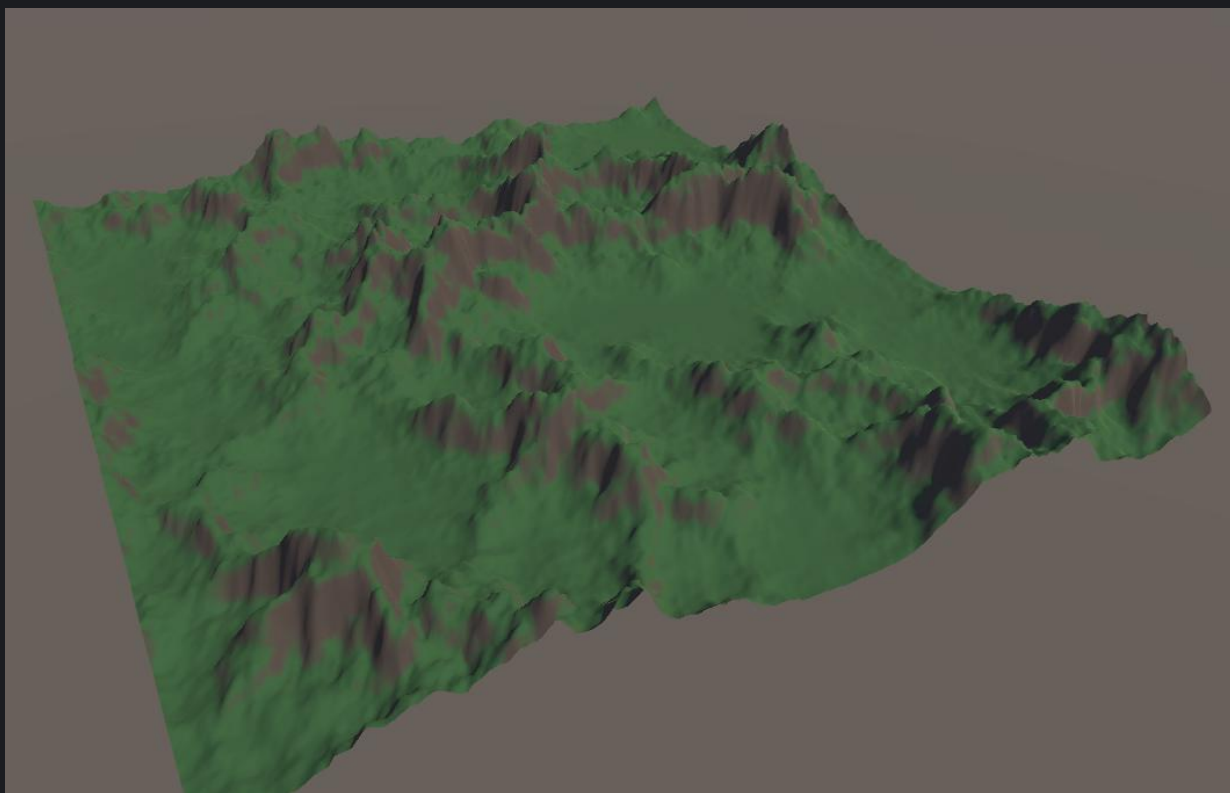
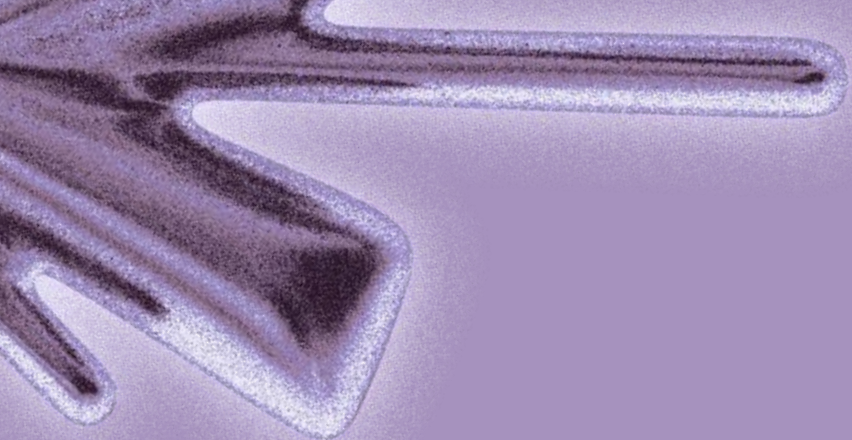
- Particle-Based Simulation
- Movement Based on Terrain
  - according to the height gradients
- Sediment Interaction
- Dynamic Sediment Capacity



- Bilinear Interpolation:
  - Sediment is distributed across nearby terrain grid points
- Weighted Brush Effect: erosion is applied gradually.
- Refinement Parameters:
  - Inertia, friction, and evaporation.



# Results



# Conclusion & Future Work

**Major Achievement:** Implemented working terrain generation tool with pseudorandom noise and erosion effects that generate realistic terrain.

**Future Improvements:**

- Exporting the generated mesh
- More interface for adjustment of erosion parameters.
- Support for more types of noise and erosion
- Optimization



Q&A